

What if the mountains won't get out of the way?

Molodensky, the telluroid, the quasigeoid and normal heights — four classes

Gravimetric Network Simulator

What we will see

1. Where we left off
2. The mountain in the way
3. The height that lives inside the mountain
4. Molodensky's big idea
5. From effort to meters
6. The telluroid
7. The quasigeoid
8. Molodensky's problem
9. The solution: Stokes plus corrections
10. With GNSS, the surface is known
11. Heights without leveling
12. Brazil and the world
13. From the quasigeoid to the IHRs
14. Wrap-up

Four classes

Class 1 · The mountain in the way

Why Stokes' formula trips over the continents — and what the density of rocks has to do with it.

Class 3 · Molodensky's problem

How to compute ζ from measurements made on the surface: Stokes, plus corrections.

Class 2 · A height without density

Normal height, telluroid, height anomaly ζ and quasigeoid.

Class 4 · In practice

GNSS + quasigeoid = normal height.
Brazil, the world and the IHRs.

Each class is 50 minutes long, starts by recalling the previous one and ends with a summary. We will go **slowly**: one idea per slide.

Part 1

The mountain in the way

class 1 of 4

Where we left off

Today's questions

1. Where does the geoid run when there is a continent on top of it?
2. Why doesn't Stokes' formula like mountains?
3. Why does the height "above sea level" need to know what the mountain is made of?
4. What was Molodensky's idea?

Deal

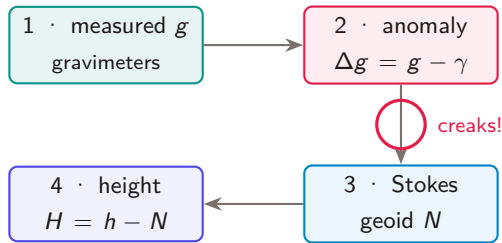
By the end of the class you will be able to answer all four — in your own words.

They come back on the last slide.

Last class's chain

In the gravimetry class we built a chain with four links:

1. measure gravity g ;
2. subtract the gravity of the smooth Earth: $\Delta g = g - \gamma$;
3. add up the anomalies of the whole planet with Stokes, and find the geoid N ;
4. with GNSS, find the height:
 $H = h - N$.



Today we will pull on one of these links
— and hear it **creak**.

Stokes' promise

Stokes' problem, in one line

In: Δg over the whole Earth.

Out: N anywhere.

A beautiful formula, from 1849. But it came with **four requirements**.

Last class ended by pointing at the first one.

For the formula to hold

1. no mass above the geoid;
2. Δg known over the whole Earth;
3. the Earth treated as almost a sphere;
4. an ellipsoid with the same mass and the same potential as the Earth.

Today: requirement number 1.

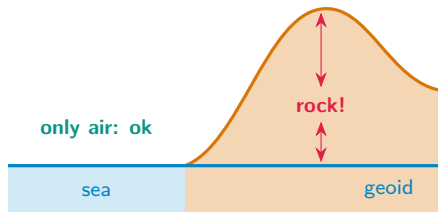
The mountain in the way

A sentence that looks harmless

“No mass above
the geoid.”

At **sea**, fine: the water surface lies practically on the geoid, and above it there is only air.

On **dry land**, no: there is rock above the geoid. Sometimes kilometers of rock.

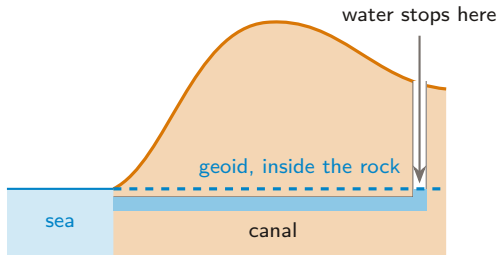


Where does the geoid run under a continent?

Imagine digging a **very thin canal** from the sea to the middle of the continent.

Sea water flows into the canal and stops at a certain level. That level is the **geoid**.

So, under high ground, the geoid runs **inside the rock**.

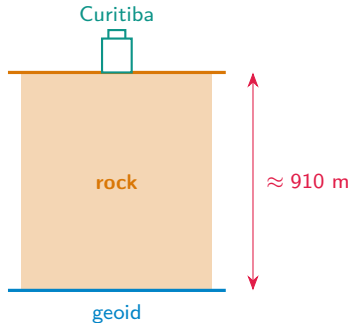


Curitiba: 900 meters of rock above the geoid

The absolute gravity station in Curitiba — the one from last class — sits at about **910 m** of height.

So between the gravimeter and the geoid there are about **900 meters of rock**.

- Brasília: about 1 100 m of rock;
- the Andes: 4 to 6 km;
- Everest: almost 9 km.

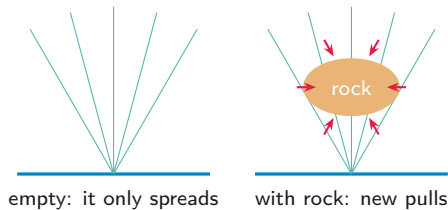


Why Stokes wants empty space

Remember the **spray paint**? The pull of gravity spreads through space the way paint spreads over the wall.

Stokes counts on that: from the geoid up, the effect of the masses **only spreads out** — no new pull appears on the way.

If there is rock in between, each piece of it is a **new pull** that the formula cannot account for.



Stop and think

The formula wants **empty** space above the geoid. The mountain **is there**.
What would you do?

(a) Pretend

Pretend the mountain does not exist and do the sum anyway.

(b) Remove

Remove the mountain — on paper — and compute how much it was pulling.

(c) Squash

Squash the mountain into a very thin sheet lying on the geoid.

Geodesists have done all three. And all three run into the same problem.

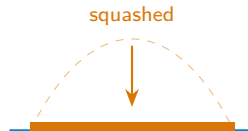
The three ways out have names



(a) free-air



(b) Bouguer



(c) Helmert condensation

The common problem

To remove or squash a mountain — and to know how much “pretending” costs — you need to know **how much it weighs**.

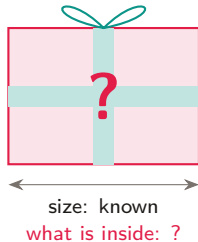
How much does a mountain weigh?

The **size** of the mountain we know: the relief is measured by satellites and radar, meter by meter.

But weight is not just size:

$$\text{mass} = \text{volume} \times \text{density}$$

And the density of the rock **deep inside**, nobody has measured.



It is like guessing the weight of a **wrapped present**.

Different rocks, different weights

material	kg/m ³
water	1 000
rock salt	≈ 2 220
sandstone	≈ 2 350
limestone	≈ 2 550
granite	≈ 2 640
basalt	≈ 2 990
standard in reductions	2 670

Averages from Telford et al. (1990). Each rock varies a lot around them.

The standard value is an educated guess

2 670 kg/m³ is a typical average for the crust. Real rock, at any given place, may be 10 % above or below.

Close to home

The third plateau of Paraná — the western half of the state — is made of stacked **basalt flows**: much heavier than average.

The price of guessing the density

The Curitiba station, with last class's Bouguer plate (910 m of rock), computed with three densities:

density	plate	Bouguer anomaly
2 350	89.7 mGal	−35.6 mGal
2 670	101.9 mGal	−47.8 mGal
3 000	114.5 mGal	−60.4 mGal

The size of the guess

Just because of the density, the number changes by ± 12 mGal.

An absolute gravimeter errs by a few μ Gal: the guess is thousands of times larger than the measurement error.

**The height that lives inside the
mountain**

Recall: height measured as effort

In the geoid class we saw that the “real” height is measured as **effort**: the geopotential number

$$C = W_0 - W$$

is the work, per kilogram, to climb from the geoid to the point. To turn it into meters:

$$H = \frac{C}{\bar{g}}$$

That is the **orthometric** height: the classic “height above sea level”.

And $\bar{g}$?

It is the **mean gravity along the plumb line**, from the geoid up to the point.

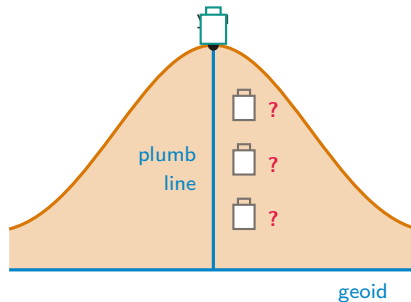
Look at the path of that plumb line. Where does it run?

A plumb line that crosses the rock

From the geoid up to you, the plumb line runs **inside the mountain**.

To know the real $\bar{g}$, you would have to measure gravity in there, meter by meter, down to the geoid.

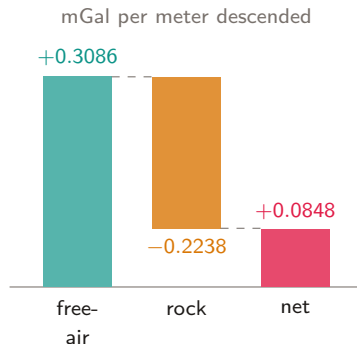
Nobody takes a gravimeter **inside** a mountain.



So it is computed — with an assumption

Go down, in your mind, from you to the geoid. **For every meter** you descend, two things happen:

1. you get closer to the center: g **increases** by 0.3086 mGal— free-air;
2. now there is rock *above* you, pulling up, and less rock *below*: g **decreases** by $2 \times 0.1119 = 0.2238$ mGal.



The Poincaré–Prey sum

Net result: g grows by **0.0848 mGal per meter** descended. Averaged over the whole path, half of that:

$$\bar{g} \approx g + 0.0424 H \quad (\text{mGal, with } H \text{ in meters})$$

Curitiba:

$$\bar{g} \approx 978\,760.4 + 0.0424 \times 910 \approx \mathbf{978\,799.0} \text{ mGal}$$

Where is the assumption?

In the 0.1119. It holds for rock of **2 670 kg/m³**.

Heavier rock, another number.
The orthometric height **inherits the guess** about the density.

A station at 2 000 m

- (a) With Prey's rule, how much *higher* than the g measured up there is $\bar{g}$?
- (b) If the mountain is basalt (2 990 kg/m³), does that number get bigger or smaller?

Your turn

A station at 2 000 m

- (a) With Prey's rule, how much *higher* than the g measured up there is $\bar{g}$?
- (b) If the mountain is basalt ($2\,990\text{ kg/m}^3$), does that number get bigger or smaller?

Answers

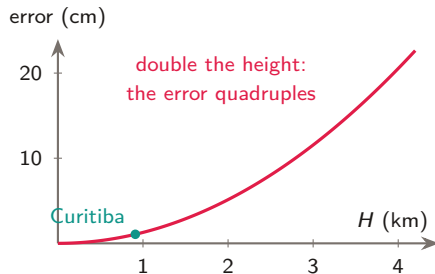
- (a) $0.0424 \times 2000 = 84.8\text{ mGal}$.
- (b) **Smaller**. Denser rock pulls up harder when you are under it: the net per meter drops from 0.0848 to 0.0578, and the sum gives about **58 mGal**.

How wrong the height gets

With the effort C fixed, $H = C/\bar{g}$.

Getting the density wrong by **300 kg/m³** (about 10 %) changes $\bar{g}$ — and, with it, H :

height	error in H
500 m	0.3 cm
1 000 m	1.3 cm
2 000 m	5.1 cm
3 000 m	11.6 cm
4 000 m	20.5 cm



In Curitiba: from 909.989 to 910.012 m, just by changing the rock.

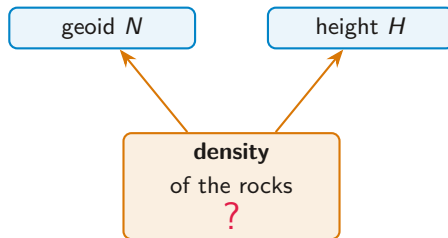
Density comes in twice

1 · In the geoid

To use Stokes, you must remove or squash the mountains: **you need the density.**

2 · In the orthometric height

$\bar{g}$ lives inside the mountain: **you need the density.**



Molodensky's question:

what if we never went inside the mountain?

Molodensky's big idea

Who Molodensky was

Mikhail Sergeevich Molodensky (1909–1991) studied at Moscow University and worked at the Soviet Union's central institute of geodesy and cartography, TsNIIGAiK.

He built the **first Soviet gravimeter**. In **1945**, in a work called *Basic problems of geodetic gravimetry*, he proposed replacing the geoid and the orthometric height with two new ideas.

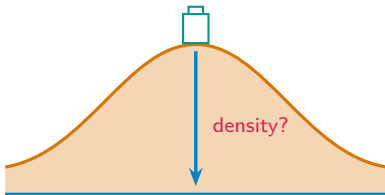
You may have seen the name

In GIS software, among the datum transformation options, there are the “**Molodensky transformations**”.

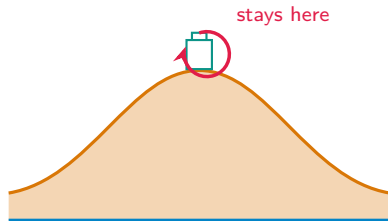
Same man.

The idea, in one sentence

Do not take the measurements **inside** the Earth.
Solve the problem **on the surface**, where the measurements are made.



Stokes



Molodensky

What changes — and what it costs

The gain

No assumption about rock density.
Only measurable things go in: gravity, leveling, position.

The price

Molodensky **does not compute the geoid**. He computes a neighboring surface, which coincides with it at sea.

Three new words are coming:

1. **normal height** H^* ;
2. **telluroid**;
3. **quasigeoid**, and its distance ζ .

One at a time, next class.

Take-home — class 1

1. Under the continents the geoid runs **inside the rock**: it is the level where water would stop in a canal connected to the sea.
2. Stokes wants **empty space** above the geoid: each piece of rock in between is a pull the formula cannot account for.
3. Removing or squashing the mountain requires knowing **how much it weighs** — and the density in there is a guess ($2\,670\text{ kg/m}^3$, $\pm 10\%$).
4. The orthometric height $H = C/\bar{g}$ depends on it too: the error grows with the **square** of the height.
5. Molodensky (1945): **solve everything on the surface**, without going into the mountain.

Next class: how to turn effort into meters without knowing what the mountain is made of — the normal height, the telluroid and the quasigeoid.

Part 2

A height that does not need the density

class 2 of 4

From effort to meters

Where we left off

In class 1:

1. under the continents, the geoid runs **inside the rock**;
2. Stokes and the orthometric height need to know **how much the rock weighs** — and that is a guess;
3. Molodensky: solve everything **on the surface**.

Today's question, in one line

How do you say “how many meters above the sea” **without knowing** what the mountain is made of?

Today's questions

1. How do you turn effort into meters without going into the mountain?
2. What is the **telluroid**?
3. What is the **height anomaly** ζ ?
4. What is the **quasigeoid** — and how far is it from the geoid?

Slowly

Three new words in a single class. We will take them one at a time, always with a drawing.

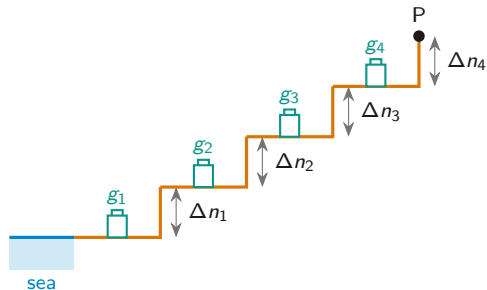
The effort C is measured

Start at the tide gauge, where the geoid touches the sea, and climb one step at a time:

- the **level and staff** measure how much each step rises: Δn ;
- the **gravimeter** measures how hard the Earth pulls there: g ;
- the effort of each step is $g \times \Delta n$.

Adding up all the steps:

$$C = g_1 \Delta n_1 + g_2 \Delta n_2 + g_3 \Delta n_3 + \dots$$



Only level, staff and gravimeter. **No rock, no density:** C is a measurement.

From effort to meters

C is energy per kilogram, in m^2/s^2 . Nobody wants a road sign written like that.

To turn it into meters, divide C by an “effort per meter” — a **gravity**:

$$\text{meters} = \frac{\text{effort}}{\text{effort per meter}} \quad \frac{\text{m}^2/\text{s}^2}{\text{m}/\text{s}^2} = \text{m}$$

The whole question is this

Which gravity?

The answer from class 1 — the mean gravity *inside the rock* — brings the density back. We need another one.

An analogy: fuel

You know **how many liters** you used on a trip. How many kilometers did you drive?

It depends on the **fuel consumption**: liters per kilometer.

$$\text{km} = \frac{\text{liters}}{\text{liters per km}}$$

Here, the liters are the effort C ; the kilometers are the height; and the consumption is gravity.

Consumption 1 · the real one

Measured on the road. But the road went **through a mountain** nobody can drive into.
⇒ **orthometric** height.

Consumption 2 · the table value

The manufacturer's table: a number **known exactly**, without going anywhere. ⇒ **normal** height.

The two rulers

	orthometric	normal
formula	$H = C/\bar{g}$	$H^* = C/\bar{\gamma}$
the gravity	the <i>real</i> one, averaged along the plumb line, inside the rock	the <i>normal</i> one, of the pretend Earth, averaged along the vertical
where it comes from	measurement + assumption	ellipsoid formula (GRS80)
needs density?	yes	no

The same C . **Only the ruler changes** that translates it into meters.

Normal gravity depends on two things:

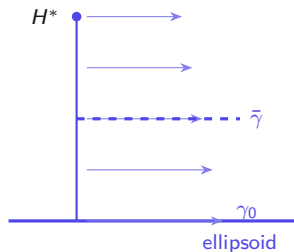
1. the **latitude** — the formula from the gravimetry class gives γ_0 , on the ellipsoid;
2. the **height** — going up, it drops by 0.3086 mGal per meter, the same old free-air.

The average, from the ellipsoid up to H^* , sits halfway:

$$\bar{\gamma} \approx \gamma_0 - 0.1543 H^*$$

(mGal, H^* in meters)

the arrow shortens going up



No rock in the sum. A calculator is enough.

Example: Curitiba

effort C (illustrative)	$8\,907.07\text{ m}^2/\text{s}^2$
γ_0 at $\varphi = -25^\circ 27'$	$978\,987.2\text{ mGal}$
$\bar{\gamma} \approx \gamma_0 - 0.1543 \times 910$	$978\,846.7\text{ mGal}$
$H^* = C/\bar{\gamma}$	909.956 m
$\bar{g}$ (Prey, $2\,670\text{ kg/m}^3$)	$978\,799.0\text{ mGal}$
$H = C/\bar{g}$	910.000 m

C of a point at 910 m orthometric height, with the g of the Curitiba station.

What if the rock were different?

From $2\,350$ to $3\,000\text{ kg/m}^3$:

H : from 909.989 to 910.012 m

H^* : **909.956 m**, always.

The difference between the two is **4.4 cm**. Keep that number in mind: it comes back shortly.

A point at latitude 45°

Given: $\gamma_0 = 980\,619.9 \text{ mGal}$ and $C = 9\,804.66 \text{ m}^2/\text{s}^2$.

- (a) Assuming $H^* \approx 1\,000 \text{ m}$, what is $\bar{\gamma}$?
- (b) What is H^* ?

Your turn

A point at latitude 45°

Given: $\gamma_0 = 980\,619.9 \text{ mGal}$ and $C = 9\,804.66 \text{ m}^2/\text{s}^2$.

(a) Assuming $H^* \approx 1\,000 \text{ m}$, what is $\bar{\gamma}$?

(b) What is H^* ?

Answers

(a) $980\,619.9 - 0.1543 \times 1000 = 980\,465.6 \text{ mGal}$, that is, 9.804656 m/s^2 .

(b) $H^* = 9\,804.66 / 9.804656 \approx 1\,000.00 \text{ m}$.

Notice: we used a *guessed* H^* to find $\bar{\gamma}$ — and the result confirmed the guess.

That is how it is done: **guess, compute, check.**

The telluroid

Who is who in the drawings

	terrain — the real Earth
	telluroid
	quasigeoid
	geoid
	ellipsoid

h	GNSS height (measured)
H^*	normal height
H	orthometric height
ζ	height anomaly
N	geoid undulation

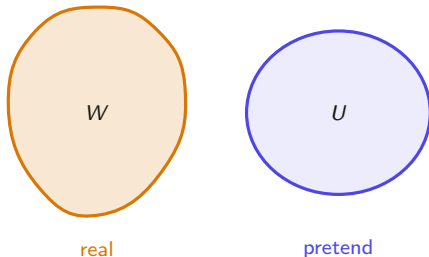
Some have not appeared yet — they will today. Each one keeps the **same color** until the end of the four classes.

Two Earths, again

Remember the gravimetry class?

- the **real Earth** has potential W — the “energy level” of each place;
- the **pretend Earth** — the ellipsoid with the Earth’s mass and rotation — has potential U , which **can be computed exactly**.

And it is agreed that both have the same “zero”: $U_0 = W_0$.



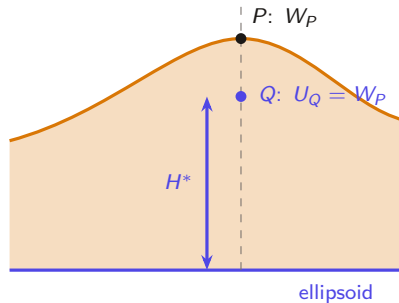
Every point has a twin

You are at P , on the real Earth. Your energy level is W_P .

Now, on the pretend Earth, move along the same vertical — down or up — until you find the point where the energy is **the same**:

$$U_Q = W_P$$

That point Q is the **twin** of P . And the height of Q above the ellipsoid is the **normal height** H^* .



Two ways of saying the same thing

Way 1 · the sum

$$H^* = C/\bar{\gamma}$$

Way 2 · the drawing

H^* = height of the twin Q above the ellipsoid

Why are they the same? Follow the steps:

1. climbing from the ellipsoid to Q , on the pretend Earth, costs $\bar{\gamma} \cdot H^*$;
2. that cost is $U_0 - U_Q$;
3. but $U_0 = W_0$ and $U_Q = W_P$, so the cost is $W_0 - W_P = C$;
4. therefore $\bar{\gamma} \cdot H^* = C$.

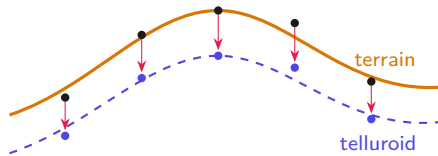
The drawing and the sum say the same thing.

Putting all the twins together: the telluroid

Find the twin of **every point** of the terrain. Together, they form a surface very much like the terrain, just shifted by a few meters: the **telluroid**.

Where the name comes from

Tellus is Latin for “earth”, “ground”; *-oid* means “like”. Telluroid: **ground-like**. Just like geoid (Earth-like) and ellipsoid (ellipse-like).



each P and its Q

ellipsoid

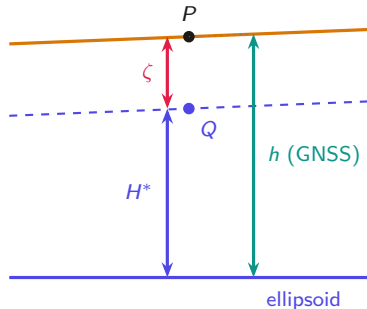
The distance between P and its twin: ζ

The distance from Q to P has a name:
height anomaly, with the Greek letter ζ
("zeta").

From the ellipsoid up to P is the GNSS
height. It is the sum of the two:

$$h = H^* + \zeta$$

ζ is about the size of the geoid undulation:
tens of meters, up or down. Negative
means Q lies above P .



Why “height anomaly”?

In geodesy, an **anomaly** is always the same idea: the real minus the normal.

gravity $\Delta g = g - \gamma$

potential $T = W - U$

height $\zeta = h - H^*$

Reading the last row

h is the **real** height of P , which GNSS measures.

H^* is the height P would have if the Earth were the **normal** one, with the same effort.

ζ measures how much the real Earth “fools” the pretend one, right there.

The quasigeoid

Swapping the order of the terms

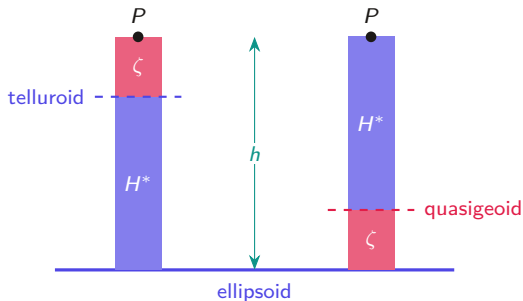
At school: $3 + 5 = 5 + 3$. The order of the terms does not change the sum.

So also

$$h = H^* + \zeta = \zeta + H^*$$

You can stack them the other way round: **first** ζ , from the ellipsoid;
then H^* , up to P .

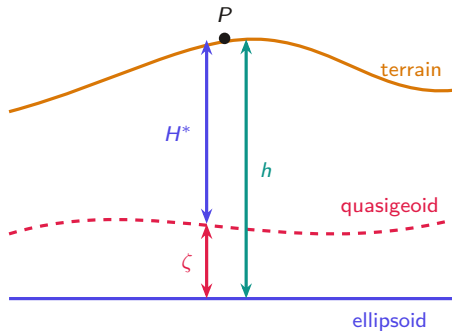
The point where the terms meet has moved.



The quasigeoid

Draw ζ upwards from the ellipsoid, **everywhere**. The surface that appears is the **quasigeoid**.

The normal height H^* is the distance from the quasigeoid up to you —
— just as the orthometric H is the distance from the *geoid* up to you.



Twin formulas

	Stokes (1849)	Molodensky (1945)
the formula	$h = H + N$	$h = H^* + \zeta$
the height	orthometric H	normal H^*
the separation	undulation N	height anomaly ζ
the reference surface	geoid	quasigeoid
is it a level surface?	yes	no
needs density?	yes	no

The same shape, $h = \text{height} + \text{separation}$. What changes is **where** you count from.

Careful: the quasigeoid is not a level surface

Water does not see it

Along the quasigeoid, the energy level W is not constant. A marble placed there may roll.

It is a **computational surface**: it exists on paper, like the Equator line or the border of a time zone.

But at sea...

At the sea surface, $C = 0$, so $H = H^* = 0$. There,

$$N = \zeta$$

The quasigeoid **coincides with the geoid** over the oceans. Hence the name: it is *almost* the geoid.

How far apart are the geoid and the quasigeoid? (1)

The two twin formulas give the same h :

$$H + N = H^* + \zeta \implies N - \zeta = H^* - H$$

And both heights come from the same C :

$$H^* - H = \frac{C}{\bar{\gamma}} - \frac{C}{\bar{g}} = \frac{C}{\bar{g}} \cdot \frac{\bar{g} - \bar{\gamma}}{\bar{\gamma}} = H \frac{\bar{g} - \bar{\gamma}}{\bar{\gamma}}$$

In words

The distance between the geoid and the quasigeoid depends on **how much the mean real gravity differs from the mean normal gravity**, times the height.

At sea, $H = 0$: zero distance.

How far apart are the geoid and the quasigeoid? (2)

Put the two averages side by side:

$$\bar{g} \approx g + 0.0424 H$$

$$\bar{\gamma} \approx \gamma_0 - 0.1543 H$$

$$\bar{g} - \bar{\gamma} \approx g - \gamma_0 + 0.1967 H$$

And $0.1967 = 0.3086 - 0.1119$: free-air minus Bouguer. That is exactly the **Bouguer anomaly** from the gravimetry class!

The result

$$N - \zeta = H^* - H \approx \frac{\Delta g_B \cdot H}{\bar{\gamma}}$$

Curitiba: $\Delta g_B = -47.8 \text{ mGal}$, $H = 910 \text{ m}$:

$$N - \zeta \approx \frac{-0.000478 \times 910}{9.79} \approx -4.4 \text{ cm}$$

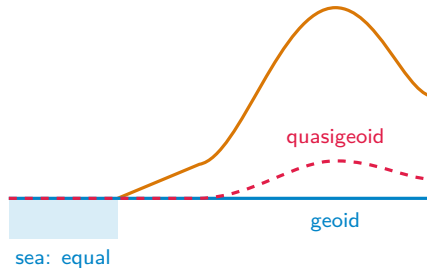
The same 4.4 cm as in the Curitiba example.

The numbers

	Δg_B (mGal)	H (m)	$N - \zeta$
sea	—	0	0
Curitiba	-47.8	910	-4.4 cm
	-100	1 000	-10 cm
	-200	2 000	-41 cm
	-300	3 000	-92 cm
	-400	5 000	-2.0 m

The last four rows are illustrative, with the simple formula.

under the mountain the quasigeoid runs above the geoid (exaggerated)



On the plains: millimeters to centimeters. In the great mountain ranges: up to meters.

Stop and think

Under big mountains the Bouguer anomaly is **negative** — those are the *roots* from the gravimetry class.

- (a) Is $N - \zeta$ positive or negative?
- (b) Does the quasigeoid run above or below the geoid?
- (c) Is H^* larger or smaller than H ?

Stop and think

Under big mountains the Bouguer anomaly is **negative** — those are the *roots* from the gravimetry class.

- (a) Is $N - \zeta$ positive or negative?
- (b) Does the quasigeoid run above or below the geoid?
- (c) Is H^* larger or smaller than H ?

Answers

- (a) **Negative**: $\Delta g_B < 0$ and $H > 0$.
- (b) **Above**: $\zeta > N$.
- (c) **Smaller**: $H^* < H$ — as in Curitiba, 4.4 cm less.

Neither one is perfect

Remember the **Great Lakes**, in the geoid class? Still water is a level surface: the same C everywhere. Even so:

- H changes from one end to the other, because $\bar{g}$ changes;
- H^* changes too, because $\bar{\gamma}$ changes with latitude;
- only the **dynamic** height — C divided by a constant — stays the same.

Each height has its advantage

The normal height's: it comes **only from measurements and a formula**, with no assumption at all about the Earth's interior.

That is what Molodensky was after.

Take-home — class 2

1. The effort C is measured with level, staff and gravimeter. To turn it into meters you divide by a gravity: **which** gravity is the question.
2. **Normal height** $H^* = C/\bar{\gamma}$: the pretend Earth's gravity, which can be computed. **No density**.
3. The **telluroid** gathers the twins Q of the terrain points: $U_Q = W_P$.
4. The **height anomaly** ζ is the distance from Q to P , and $h = H^* + \zeta$.
5. The **quasigeoid** is ζ drawn up from the ellipsoid. It coincides with the geoid at sea; in the mountains, $N - \zeta \approx \Delta g_B H / \bar{\gamma}$.

Try it: in the **Earth Models** simulator, **Telluroid** tab. **Next class:** how ζ is computed — Molodensky's problem.

Part 3

Molodensky's problem

class 3 of 4

Molodensky's problem

In class 2:

1. normal height $H^* = C/\bar{\gamma}$: **no density**;
2. telluroid: the twins Q , with $U_Q = W_P$;
3. $h = H^* + \zeta$, and the quasigeoid is ζ drawn up from the ellipsoid.

One piece is missing

Where there is GNSS *and* leveling, $\zeta = h - H^*$ comes for free. But such points are few.

To have ζ **everywhere**, you have to compute it — from gravity.

Today's questions

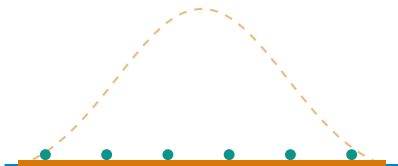
1. What, exactly, is the problem Molodensky solved?
2. Which gravity anomaly does he use — and does it need the density?
3. What is left of Stokes' formula?
4. What changed when GNSS arrived?

Formulas today

Some will look scary. No need to memorize them: we will **read each one aloud**, piece by piece.

Same question, different stage

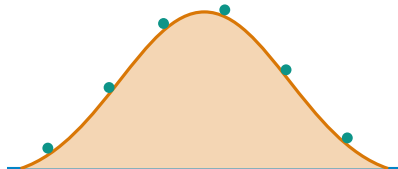
Stokes



Stokes

Data **on the geoid** (after pushing the mountains away) $\Rightarrow$ finds the **geoid**.

Molodensky



Molodensky

Data **on the surface**, where they were measured $\Rightarrow$ finds the **surface itself** and the gravity field outside it.

A strange problem: the boundary is the unknown

The measurements lie on the Earth's surface. But the shape of that surface — in the gravity-field sense — is part of what we want to find out.

It is like drawing the map of an island with measurements made **on the very coast** you are drawing.

The technical name

Free boundary-value problem: the boundary carrying the data is also unknown.

Molodensky's way out

Start from a **good draft** of the surface and compute only the **correction**.

The adjustment trick: approximate + correction

Anyone who does least-squares adjustment knows this trick:

$$X = X_0 + x$$

an **approximate value** X_0 , which is known, plus a **small correction** x , which is computed.

Because the correction is small, the equations become **linear**. Molodensky did exactly that.

	adjustment	Molodensky
approximate	X_0	telluroid, U
correction	x	ζ , T
true	X	surface, W

$$\text{surface} = \text{telluroid} + \zeta$$

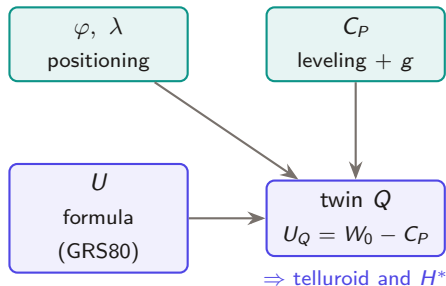
$$W = U + T$$

The draft can be computed

For each point P :

1. φ and λ , from positioning: they say **which vertical**;
2. C_P , from leveling with gravity;
3. $W_P = W_0 - C_P$;
4. on that vertical, look for the height where $U = W_P$: that is the twin Q .

Nothing was guessed: the telluroid is **known before we start**.



What is measured at each point

what	with what	what for
g_P	gravimeter	the anomaly
C_P	level, staff and gravimeter	the twin Q and H^*
φ, λ	positioning	where the vertical is
relief	satellite and radar	a correction (coming soon)
U, γ	ellipsoid formula	the pretend Earth

Look for the word “density” in this table. It is not there.

Molodensky's anomaly

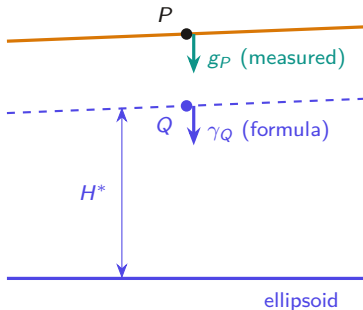
$$\Delta g = g_P - \gamma_Q$$

The gravity **measured at P** minus the **normal gravity at the twin Q** .

Since Q sits at a height H^* above the ellipsoid:

$$\gamma_Q \approx \gamma_0 - 0.3086 H^*$$

It is the **free-air anomaly** from the gravimetry class. But now it **lives on the surface**, where it was measured. Nobody went down, nobody removed a mountain.



Example: Curitiba, again

	mGal
g_P (RENEGA)	978 760.4
γ_0 at $\varphi = -25^\circ 27'$	978 987.2
-0.3086×909.96	-280.8
γ_Q	978 706.4
$\Delta g = g_P - \gamma_Q$	+54.0

Compare with class 1

Curitiba's Bouguer anomaly changed by ± 12 mGal depending on the rock chosen.

This one does not: **+54.0 mGal for any rock**. There is nothing to guess.

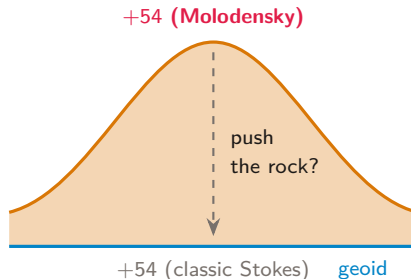
Two readings of the same number

Stokes, the classic way

+54 mGal is treated as if it were **on the geoid**, 900 m below. For that, the mountain has to be pushed inside — with an assumed density.

Molodensky

+54 mGal stays **where it was measured**, at 910 m. It is the formula that adapts to the surface.



A mountain station

Latitude 45° , $\gamma_0 = 980\,619.9$ mGal.

Normal height $H^* = 2\,000$ m. Mea-

sured gravity: $g_P = 980\,052.7$ mGal.

(a) What is γ_Q ?

(b) What is Molodensky's anomaly?

Your turn

A mountain station

Latitude 45° , $\gamma_0 = 980\,619.9$ mGal.
Normal height $H^* = 2\,000$ m. Measured gravity: $g_P = 980\,052.7$ mGal.

- (a) What is γ_Q ?
- (b) What is Molodensky's anomaly?

Answers

(a) $980\,619.9 - 0.3086 \times 2\,000 = 980\,002.7$ mGal.

(b)
 $980\,052.7 - 980\,002.7 = +50.0$ mGal.

With the full normal-gravity formula (appendix) you get +49.6: the 0.3086 rule is a very good approximation, not the exact sum.

Bruns, on the surface

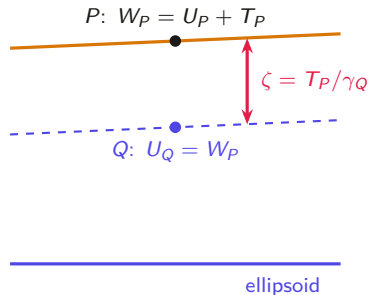
In the gravimetry class, Bruns linked energy to meters **on the geoid**:

$$N = \frac{T}{\gamma}$$

Molodensky uses the same idea **at the point P** :

$$\zeta = \frac{T_P}{\gamma_Q}$$

At P , the real Earth has a little more energy than the pretend one: T_P . “Spending” that excess by climbing against γ takes exactly ζ meters.



The fundamental equation, on the surface

The same rule as in the gravimetry class, now holding at every point of the surface:

$$\Delta g = -\frac{\partial T}{\partial h} + \frac{1}{\gamma} \frac{\partial \gamma}{\partial h} T \approx -\frac{\partial T}{\partial r} - \frac{2}{r} T$$

In words: the anomaly is **how much T changes as you go up**, plus a little piece of T itself. Same tools; **different stage**.

And requirement no. 1?

Above the Earth's surface there is only air. For Molodensky, “no mass outside the boundary” is **simply true** — because the boundary is the ground itself.

**The solution: Stokes plus
corrections**

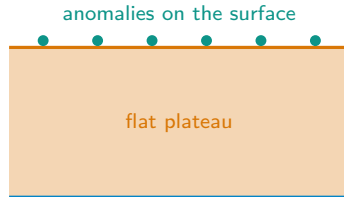
If the ground were flat

Imagine a **perfectly flat** plateau, all at the same height.

In that case, Molodensky's solution is exactly Stokes' formula, applied to the **surface** anomalies:

$$\zeta = \frac{R}{4\pi\gamma} \iint_{\sigma} \Delta g S(\psi) d\sigma$$

Same function $S(\psi)$, same “every little piece votes”. Only the anomalies and the result changed: ζ instead of N .



Stokes **was not thrown away**: he is the first step of Molodensky's solution.

But the ground has hills: the correction G_1

$$\zeta \approx \frac{R}{4\pi\gamma} \iint_{\sigma} (\Delta g + G_1) S(\psi) d\sigma$$

$$G_1 = \frac{R^2}{2\pi} \iint_{\sigma} \frac{h - h_P}{\ell_0^3} \Delta g d\sigma$$

Reading G_1 aloud. For **each neighbor**:

1. how far above or below you it is: $h - h_P$;
2. times its anomaly: Δg ;
3. divided by the distance **cubed**: ℓ_0^3 .

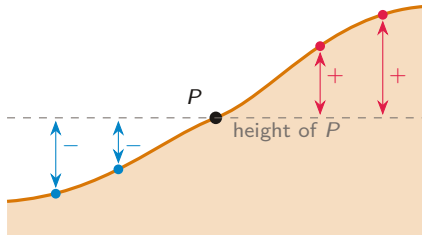
Add it all up. Done: it is a correction **added to the anomalies** before Stokes.

No need to memorize

What matters is the idea: G_1 **measures how far the surrounding terrain is from flat**, and corrects the anomalies for it.

Reading G_1 slowly

- **Flat** terrain around: $h - h_P = 0$ for every neighbor, $G_1 = 0$, and pure Stokes comes back.
- **Higher** neighbors enter with a positive $h - h_P$; **lower** ones, with a negative $h - h_P$.
- The **steeper** the terrain, the larger the correction.



Cubed: only the neighborhood matters

distance	$1/\ell^2$	$1/\ell^3$
1	1	1
2	1/4	1/8
3	1/9	1/27
10	1/100	1/1 000

Remember the **spray paint**, which fell off “squared”? Cubed falls off even faster.

A neighbor 10 times farther away weighs **a thousand times less**.

That is why G_1 is computed from the relief and the gravity data **close to** each point.

G_1 looks like an old friend

In the gravimetry class, the **terrain correction** corrected gravity for the hills and valleys around the station.

When the terrain is only gently curved and the anomaly grows with height the way the Bouguer model predicts, G_1 becomes **almost equal** to that terrain correction.

Similar, not the same

The resemblance is in the shape. G_1 is computed from the **measured anomalies** and the **relief** — not from an assumed density.

It is the kind of thing that makes a geodesist smile: two different theories arriving at the same picture.

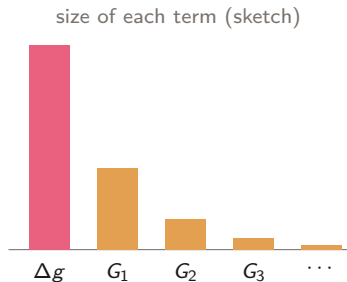
A series: each term smaller than the last

Molodensky's full solution is a **series**:

$$\zeta = \frac{R}{4\pi\gamma} \iint (\Delta g + G_1 + G_2 + G_3 + \dots) S(\psi) d\sigma$$

Each term is much smaller than the previous one — like $0.3 + 0.03 + 0.003 + \dots$

In practice, **the first term is enough**, sometimes the second.



On very steep relief the series misbehaves. Practical fix: smooth the relief first (appendix).

What each theory needs

	Stokes	Molodensky
where the anomalies are	on the geoid	on the surface
what happens to the relief	remove or squash	goes into G_1
rock density	needed	not needed
what comes out	N , geoid	ζ , quasigeoid
matching height	orthometric	normal
when	1849	1945

The central tool — adding up anomalies weighted by $S(\psi)$ — is **the same in both**.

What if I still want the geoid?

You can. Compute ζ with Molodensky and add the separation from class 2:

$$N = \zeta + (N - \zeta) \approx \zeta + \frac{\Delta g_B H}{\bar{\gamma}}$$

But look at Δg_B : it is the **Bouguer** anomaly — and it has the density inside.

The density comes back...

... only if you insist on the geoid.

Molodensky does not forbid the geoid. He just shows that **you can live without it** — and measure heights with no guess at all.

With GNSS, the surface is known

1945 and today

1945

The vertical position of a point came from **leveling**. The shape of the surface, in the field sense, was unknown.

Free boundary-value problem: hence the telluroid.

Today

GNSS gives h directly, in minutes, anywhere. The surface is known **geometrically**.

Fixed boundary-value problem: the boundary is already in place.

Molodensky's idea — solve on the surface — became **even more natural** with GNSS.

The gravity disturbance

If GNSS tells where P is, g and γ can be compared **at the same point**:

$$\boxed{\delta g = g_P - \gamma_P} \quad (\text{and not } \Delta g = g_P - \gamma_Q)$$

The difference between them is the difference between γ at P and at Q , which are ζ meters apart:

$$\Delta g - \delta g \approx -0.3086 \zeta$$

With $\zeta = -5$ m: **+1.5 mGal**.

Two versions of the same idea

Δg is the natural choice when height comes from **leveling**: Molodensky's.

δg is the natural choice when height comes from **GNSS**.

Neither one asks for density. Today's models use both.

Take-home — class 3

1. Molodensky solves the problem **on the surface**: the data stay where they were measured, and the surface is part of the answer.
2. As in adjustment: **approximate + correction**. Surface = telluroid + ζ ; $W = U + T$.
3. Molodensky's anomaly: $\Delta g = g_P - \gamma_Q$ — the free-air one, on the surface. In Curitiba, +54 mGal for **any** rock.
4. Bruns on the surface: $\zeta = T_P/\gamma_Q$. Solution: **Stokes + G_1 + ...**; on flat terrain, just Stokes.
5. With GNSS the surface is known: the **disturbance** $\delta g = g_P - \gamma_P$ enters the scene.

Next class: heights without leveling — GNSS + quasigeoid, Brazil, the world and the IHRs.

Part 4

In practice: GNSS, quasigeoid and normal heights

class 4 of 4

Heights without leveling

Where we left off

In class 3:

1. Molodensky's anomaly,
 $\Delta g = g_P - \gamma_Q$, lives on the surface
and **needs no density**;
2. ζ comes from Stokes **plus corrections**
($G_1, \dots$);
3. with GNSS, the surface is known, and
the disturbance δg comes in.

Today

Use ζ for what surveyors care about:
real heights, in Brazil, today.

Today's questions

1. How do you get a height from a GNSS receiver?
2. How do you know whether a model of ζ is good?
3. Which height does Brazil use — and since when?
4. How does all this connect to the IHRF, the world height reference system?

Last class

At the end, one slide ties the four classes into a single story.

Leveling is slow

Leveling

Level and staff, step by step, there and back: **a few kilometers a day**.

IBGE has been leveling Brazil since **1945** — the year of Molodensky's theory — and there are still regions, such as parts of the Amazon, where the forest gets in the way.

GNSS

An antenna on a tripod: h in **minutes or hours**, anywhere with open sky.

But h is a **geometric** height: water does not obey it.

The bridge between the two is ζ .

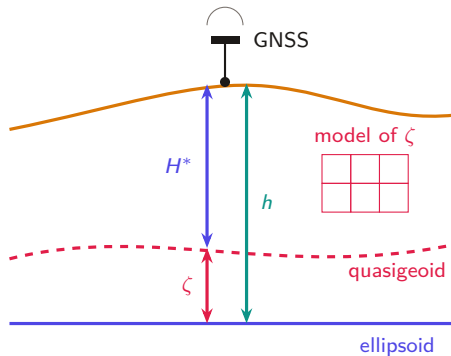
The recipe: $H^* = h - \zeta$

Three ingredients:

1. the h from **GNSS**;
2. the ζ from a **model** (a grid of values, read at the point's location);
3. a **subtraction**:

$$H^* = h - \zeta$$

This is “GNSS leveling”.



Step-by-step example

h (GNSS, SIRGAS2000)	812.47 m
------------------------	----------

ζ (model, at the point)	-3.21 m
-------------------------------	---------

$H^* = 812.47 - (-3.21)$	815.68 m
--------------------------	----------

Illustrative numbers.

Watch the sign

Negative ζ : the quasigeoid runs **below** the ellipsoid. So the normal height comes out **larger** than h .

Minus a minus is a plus — a common field mistake is exactly this one.

The uncertainty of H^* combines that of h and that of the model.

All three must speak the same language

1. h in the **same reference frame** as the model — in Brazil, SIRGAS2000, with the GRS80 ellipsoid;
2. ζ from a model made **for your height system**;
3. H^* in the **same system** as the benchmarks — in Brazil, the normal heights of REALT-2018.

Mixing spoils it

Switching the frame of h , or using a model built for another kind of height, spoils **centimeters** — sometimes **meters**.

It is the thesis of the reference-frames class: a number must say **which reference frame** it is in.

How do you know the model is right?

At points that have **both** — GNSS and leveling — the sum must close:

$$\text{residual} = h - H^*_{\text{leveled}} - \zeta_{\text{model}}$$

If the model were perfect, **zero** everywhere.

And if the residuals show a **trend**? A correction surface is fitted — by **least squares**, of course.

residuals at GNSS/leveling points



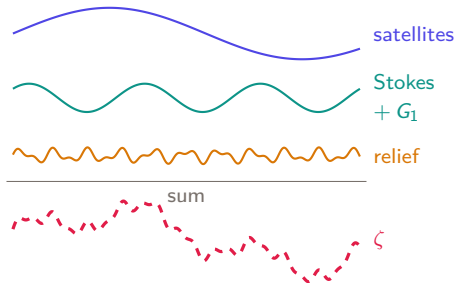
You have seen this number

MAPGEO2015: ± 0.17 m at 592
GNSS/leveling points — from the
gravimetry class.

Remove, compute, restore — Molodensky version

The same scheme as in the gravimetry class. What goes in changes: **surface** anomalies.

1. **remove**: the global satellite model (large shapes) and the effect of the nearby relief;
2. **compute**: with what is left, small and local: Stokes + G_1 by FFT, or **least-squares collocation**;
3. **restore**: add back what was removed.



Collocation: the adjustment idea extended to continuous functions, systematized by Moritz. Yes, least squares again.

The state of the art: the Colorado experiment

14 groups from 14 countries received the same data for a mountainous region of Colorado (USA), 730×560 km, and computed the geoid and the quasigeoid, each with their own method.

The referee: the **223 marks** of GSVS17, a line of top-quality GNSS and leveling.

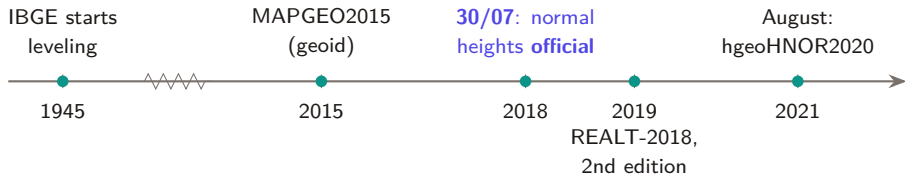
What came out

- the models agree to **~ 2 cm**;
- accuracy along the line: **1.2 to 3.4 cm**;
- the **1 cm** goal has not been reached yet — in the mountains.

Wang et al. (2021), J. Geod. 95:127.

Brazil and the world

Brazil adopted normal heights



Since **30 July 2018**, the official heights of the Brazilian Geodetic System are **normal** heights, referred to the quasigeoid.

They come from **REALT-2018**: the readjustment of the leveling network with **geopotential numbers**.

In other words: Brazil does exactly what classes 2 and 3 described — measured C , $H^* = C/\bar{\gamma}$.

What changed at the benchmarks

Before 2018, the official heights were **normal-orthometric**: leveling corrected with *normal* gravity, without measured gravity at every benchmark.

REALT-2018 redid everything with geopotential numbers and **real gravity at every benchmark**.

Only changing the type of height

With the same data: **94 %** of the benchmarks changed between -15 mm and $+5$ mm.

The whole readjustment

About **50 000** benchmarks (76 %) changed by **$+20$ to $+30$ cm**, in much of the North, Northeast, Center-West and Southeast.

The lesson: the *type* of height changes millimeters; **data, gravity and adjustment** change decimeters.

IBGE publishes, for every place in Brazil, a **conversion factor** η and its uncertainty:

$$H^* = h - \eta$$

with h in SIRGAS2000 and H^* in the normal heights of REALT-2018 — referred to the country's two vertical datums, **Imbituba** and **Santana**.

A translator, not a pure quasigeoid

The model was built to **agree with the official benchmarks**. IBGE itself warns: it is not a gravimetric model of the quasigeoid.

Compared with MAPGEO2015

Difference below 50 cm over most of the country; up to -95 cm in the Amazon. Conversion uncertainty: **less than half**.

And the world?

height	where
normal	Europe (EVRS), Germany, Russia, China, Brazil
orthometric	USA (NAVD 88), Canada (CGVD2013)
dynamic	Great Lakes (IGLD)

Europe

The European vertical reference system (EVRS) uses **normal heights** from geopotential numbers, in three realizations: EVRF2000, EVRF2007 and EVRF2019. Most European countries follow Molodensky.

There is no winner. There are **choices**, and each country must say which one it made.

From the quasigeoid to the IHRS

From ζ to the potential

The IHRS measures height as a

geopotential number: $C_P = W_0 - W_P$.

With GNSS and ζ , W_P comes out without leveling:

$$W_P = \underbrace{U_P}_{\text{formula, with } h} + \underbrace{T_P}_{\gamma \zeta \text{ (Bruns)}}$$

And then $C_P = W_0 - W_P$. With $U_0 = W_0$, this is the same as

$$C_P \approx \bar{\gamma} (h - \zeta) = \bar{\gamma} H^*$$

What this means

GNSS + quasigeoid $\Rightarrow$ **the IHRS vertical coordinate**, at any point, without leveling.

This is how IHRF stations get their potential: position in the ITRF + gravity field.

Example: from GNSS to the geopotential number

$$H^* = h - \zeta \text{ (earlier example)} \quad 815.68 \text{ m}$$

$$\bar{\gamma} \text{ at } 45^\circ \quad 980\,494.1 \text{ mGal}$$

$$C = \bar{\gamma} H^* \quad 7\,997.69 \text{ m}^2/\text{s}^2$$

$$W_P = W_0 - C \quad 62\,628\,855.71 \text{ m}^2/\text{s}^2$$

Illustrative; latitude 45° for round numbers.

$$W_0 = 62\,636\,853.4 \text{ m}^2/\text{s}^2 \text{ (IHRs)}.$$

Watch the units

$\bar{\gamma}$ in m/s^2 (9.804941) times H^* in meters gives m^2/s^2 : energy per kilogram.

It is the same effort C as in class 2 — now coming from GNSS.

The pipe, once more

In the geoid class: A and B , 30 km apart, at the same latitude, with the **same** $h = 812.00$ m. With the geoid ($N_A = -8.40$, $N_B = -8.12$), water ran 28 cm downhill from A to B .

Now, with the quasigeoid: $\zeta_A = -8.36$ m and $\zeta_B = -8.08$ m.

- (a) What are H_A^* and H_B^* ?
- (b) Which way does the water run, and how far down?

Illustrative numbers, with $\Delta g_B \approx -48$ mGal and $H \approx 820$ m.

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Answers

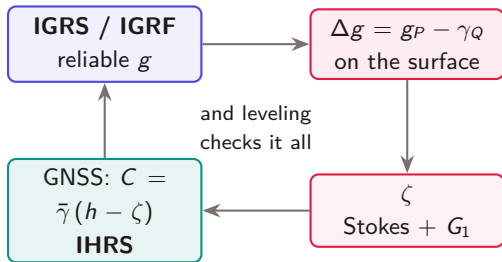
(a) $H_A^* = 812.00 + 8.36 = 820.36$ m;
 $H_B^* = 812.00 + 8.08 = 820.08$ m.

(b) From A to B , dropping **28 cm** — the same as before.

Swapping the geoid for the quasigeoid **does not change the physics**: only the ruler changes. (Each H^* ended up 4 cm below H , as in Curitiba.)

Illustrative numbers, with $\Delta g_B \approx -48$ mGal and $H \approx 820$ m.

The loop closes, Molodensky version



No density along the way

From measured gravity to the height, **everything is measurement or formula**. That is what Molodensky wanted in 1945 — and GNSS made it practical.

Wrap-up

Stokes and Molodensky: not a fight

where	$N - \zeta$
at sea	zero
on the plains	millimeters to centimeters
in the great ranges	decimeters to meters

And that difference **can be computed** — if you accept a density.

Geoid

The true level surface: wherever the physics of water needs it.

Quasigeoid

Heights from **measurements only**, with no assumption about the Earth's interior.

The four classes on one slide

Class 1 · The problem

The geoid runs inside the rock. Stokes and the orthometric height need the density — a guess.

Class 3 · The computation

Anomaly on the surface, Bruns at the point, Stokes + G_1 : out comes ζ .

Class 2 · The words

$H^* = C/\bar{\gamma}$, telluroid, ζ , quasigeoid: $h = H^* + \zeta$.

Class 4 · The practice

GNSS — ζ = normal height. Brazil since 2018. The IHRs at any point.

One idea runs through all four: **do not go into the mountain.**

Take-home — class 4

1. $H^* = h - \zeta$: GNSS plus a model of ζ gives normal height without leveling — if all three are in the same reference frame.
2. Models are tested at points with GNSS *and* leveling; trends are corrected by least squares.
3. In mountains, today's best is ~ 2 cm (Colorado). The goal is 1 cm.
4. Brazil: **normal** heights since 30 July 2018 (REALT-2018); **hgeoHNOR2020** converts h into H^* .
5. GNSS + ζ gives W_P and C_P : the **IHR**S coordinate, at any point.

Try it: in the **Earth Models** simulator, **Telluroid** tab; in the **Gravimetric Network** simulator, adjust the Paraná network — its g values are the ones that go into Δg .

Appendix

For those who want to go further

A1 · Normal gravity at height and $\bar{\gamma}$

Normal gravity at height h (second order):

$$\gamma(h) = \gamma_0 \left[1 - \frac{2}{a} (1 + f + m - 2f \sin^2 \varphi) h + \frac{3h^2}{a^2} \right]$$

with $m = \omega^2 a^2 b / GM$. The mean from 0 to H^* :

$$\bar{\gamma} = \gamma_0 \left[1 - (1 + f + m - 2f \sin^2 \varphi) \frac{H^*}{a} + \frac{H^{*2}}{a^2} \right]$$

and $H^* = C/\bar{\gamma}$ is solved by iteration (two passes are enough).

Poincaré–Prey, with density ρ :

$$\bar{g} \approx g + \left(\frac{1}{2} 0.3086 - 2\pi G \rho \right) H$$

With $\rho = 2\,670$: $\bar{g} \approx g + 0.0424 H$
(Helmert).

Sensitivity to an error $\delta\rho$:

$$\delta H \approx \frac{2\pi G \delta\rho H^2}{g}$$

GRS80 and γ_0 (Somigliana): appendix A2 of the gravimetry class.

A2 · Telluroid, quasigeoid and the normal correction

Telluroid: Q on the ellipsoidal normal through P , with $U_Q = W_P$. Then $H^* = h_Q$ and $\zeta = h_P - h_Q$.

Quasigeoid: the points ζ above the ellipsoid. It is not an equipotential surface.

Separation:

$$N - \zeta = H^* - H = \frac{\bar{g} - \bar{\gamma}}{\bar{\gamma}} H \approx \frac{\Delta g_B}{\bar{\gamma}} H$$

Normal correction of leveling between A and B , with γ_0^{45} a constant:

$$H_B^* - H_A^* = \sum \delta n + NC_{AB}$$

$$NC_{AB} = \sum \frac{g - \gamma_0^{45}}{\gamma_0^{45}} \delta n + \frac{\bar{\gamma}_A - \gamma_0^{45}}{\gamma_0^{45}} H_A^* - \frac{\bar{\gamma}_B - \gamma_0^{45}}{\gamma_0^{45}} H_B^*$$

The orthometric one has the same form, with $\bar{g}$ instead of $\bar{\gamma}$ — and that is where density comes in.

A3 · The linearized Molodensky problem

Unknowns: the surface S and the potential W outside it. Data on S : W (via C and W_0) and g . With $W = U + T$ and the telluroid as approximate surface:

$$\Delta g = g_P - \gamma_Q = -\frac{\partial T}{\partial h} + \frac{1}{\gamma} \frac{\partial \gamma}{\partial h} T, \quad \zeta = \frac{T_P}{\gamma_Q}, \quad \nabla^2 T = 0 \text{ outside } S$$

Molodensky's series:

$$\zeta = \frac{R}{4\pi\gamma} \iint_{\sigma} (\Delta g + G_1 + G_2 + \cdots) S(\psi) d\sigma$$

$$G_1 = \frac{R^2}{2\pi} \iint_{\sigma} \frac{h - h_P}{\ell_0^3} \Delta g d\sigma, \quad \ell_0 = 2R \sin \frac{\psi}{2}$$

- in G_1 , $h - h_P$ is the height difference to the neighbor: normal or ellipsoidal makes no difference;
- planar approximation and Δg linear in h : $G_1 \approx$ terrain correction;
- the series converges for slopes below 45° (Moritz); in practice the relief is smoothed or analytical continuation is used.

A4 · Fixed boundary: GNSS, disturbance and Hotine

With h from GNSS, the surface is given. The boundary condition uses the **disturbance**:

$$\delta g = g_P - \gamma_P = -\frac{\partial T}{\partial h} \approx -\frac{\partial T}{\partial r}$$

On the sphere, the solution is **Hotine's** integral:

$$T = \frac{R}{4\pi} \iint_{\sigma} \delta g \mathcal{H}(\psi) d\sigma$$
$$\mathcal{H}(\psi) = \frac{1}{\sin(\psi/2)} - \ln\left(1 + \frac{1}{\sin(\psi/2)}\right)$$

Relation between the two anomalies:

$$\Delta g = \delta g + \frac{1}{\gamma} \frac{\partial \gamma}{\partial h} T$$
$$\Delta g - \delta g \approx \frac{\partial \gamma}{\partial h} \zeta \approx -0.3086 \zeta$$

(Δg in mGal, ζ in meters).

Stokes uses Δg and the kernel $S(\psi)$; Hotine uses δg and $\mathcal{H}(\psi)$. In both, $\zeta = T/\gamma$ at the end.

A5 · The potential at a station (IHR5)

$$W_P = U(\varphi_P, h_P) + T_P, \quad T_P = \gamma_Q \zeta_P$$

$$C_P = W_0 - W_P, \quad W_0 = 62\,636\,853.4 \text{ m}^2/\text{s}^2$$

U comes in closed form from GRS80 given (φ, h) ;
 T_P comes from a global model plus local
gravimetry (remove-compute-restore).

$$U_0 \neq W_0$$

In class we used $U_0 = W_0$. But
GRS80 has

$U_0 = 62\,636\,860.850 \text{ m}^2/\text{s}^2$, and the
IHR5 adopted
 $W_0 = 62\,636\,853.4 \text{ m}^2/\text{s}^2$.

The difference, $\approx 7.45 \text{ m}^2/\text{s}^2$
($\approx 0.76 \text{ m}$), enters as a **zero-degree
term** in ζ — or one uses
 $W_P = U_P + T_P$ directly, as above.

- **Molodensky** — 16 June 1909 – 12 November 1991; Moscow University (1932), TsNIIGAiK, first Soviet gravimeter; *Basic problems of geodetic gravimetry* (1945). Molodensky, Eremeev & Yurkina, *Methods for Study of the External Gravitational Field and Figure of the Earth* (1960; transl. 1962).
- **Theory** — Heiskanen & Moritz (1967) and Hofmann-Wellenhof & Moritz (2006), *Physical Geodesy*, ch. 4 (heights) and 8 (Molodensky). Moritz (1980), *Advanced Physical Geodesy*. Moritz (1980), *Geodetic Reference System 1980*.
- **Densities** — Telford, Geldart & Sheriff (1990), *Applied Geophysics*, table 6.1 (averages).
- **Brazil** — IBGE: REALT-2018 (2nd ed. 2019); normal heights since 30 July 2018; *hgeoHNOR2020 model for converting geometric into normal heights*, Methodological Reports v. 47 (2021).
- **EVRS** — BKG, *Definition of the EVRS*; Sacher & Liebsch, *EVRF2019 as new realization of EVRS*. NAVD 88 and IGLD: NOAA/NGS.
- **Colorado** — Wang et al. (2021), *Colorado geoid computation experiment: overview and summary*, J. Geod. 95:127.
- **Curitiba** — RENEGA, as in the gravimetry class. **Derived numbers**: script `quase_geoide_numeros.py`, in this folder.

The facts and numbers in this class were checked on 5 October 2026.